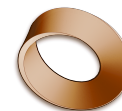


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.4 – Lenses & Image Formation

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can distinguish between converging (convex) and diverging (concave) lenses based on their shape and how they bend light.
- b) I can draw basic ray diagrams to show how a converging lens forms an image.
- c) I can explain why real images formed by converging lenses are upside down and give examples of where we use them in everyday life.

Theory

Lenses are transparent objects, usually made of glass or plastic, that bend light rays as they pass through. The most common type is the **converging lens**, also known as a **convex lens**. This lens is thicker in the middle than at the edges. When parallel rays of light travel through a converging lens, they are refracted (bent) inwards so that they meet at a single point called the **principal focus** or **focal point**. The distance from the centre of the lens to this focal point is called the **focal length**.

In contrast, a **diverging lens**, also known as a **concave lens**, is thinner in the middle than at the edges. When parallel rays of light pass through a diverging lens, they are bent outwards, spreading apart as if they originated from a focal point behind the lens. Diverging lenses always produce images that are smaller and upright, which is why they are often used to correct short-sightedness in glasses.

To understand how converging lenses form images, we use **ray diagrams**. These are simple drawings that track the path of light. There are three key rules for drawing rays through a thin converging lens: 1. A ray travelling parallel to the axis is refracted through the focal point on the other side. 2. A ray passing through the centre of the lens continues in a straight line without bending. 3. A ray passing through the focal point before reaching the lens emerges parallel to the axis.

When an object is placed further away from a converging lens than its focal length (for example, a distant tree or building), the light rays cross over at the focal point and form a **real image**. A real image can be caught on a screen, such as a piece of white paper. Because the rays have crossed, the image is always **inverted** (upside down) and often smaller than the object if the object is very far away. This principle is used in cameras, where the lens focuses light onto a sensor or film to create an upside-down picture that the camera then flips right-side up for you to see.

Word bank: convex · concave · focal point · real image · inverted

Activity

Activity: Mapping the Path of Light

You will need a piece of paper, a pencil, a ruler, and a drawing of a converging lens (a vertical line with outward-pointing arrows at each end like this: >—<).

- Draw a straight horizontal line across the middle of your paper. This is the **principal axis**.
- Mark the centre of your lens on this line and label it **C** (Centre).
- Mark two points equidistant from C on either side of the lens. Label them **F** (Focus). The distance CF is your focal length.

- Draw a tall arrow (the object) to the left of the lens, placed further away than F. Label the top of the arrow **A**.
- Using a ruler and pencil, draw two rays from point A:
- Ray 1: Draw a line parallel to the axis until it hits the lens, then draw a line through the focal point **F** on the right side.
- Ray 2: Draw a straight line from A through the centre **C** of the lens without bending.
- Where these two lines cross on the right side of the lens, mark the top of the image. Draw the base of the image down to the axis.

Observation Table: Complete the table below based on your diagram and what you know about real images.

Feature	Description of Image Formed by Converging Lens (Object > Focal Length)
Orientation	(Upright or Upside down?)
Type	(Real or Virtual?)
Size	(Larger, smaller, or same size as object? *Hint: depends on distance)
Can it be caught on a screen?	(Yes or No?)

Questions

1. Why is a converging lens also called a convex lens? Describe its shape in your own words.

2. In a ray diagram, what happens to a light ray that passes through the exact centre of a thin converging lens?

3. Explain why the image formed on the screen in a cinema projector or a camera is upside down. Use the concept of light rays crossing at the focal point in your answer.

4. If you looked through a diverging (concave) lens at a distant building, would the building appear larger or smaller than it actually is? Why?

Extension

Find out how the human eye uses a converging lens to see clearly. What part of the eye acts as the lens, and what acts as the screen where the image is formed?
